

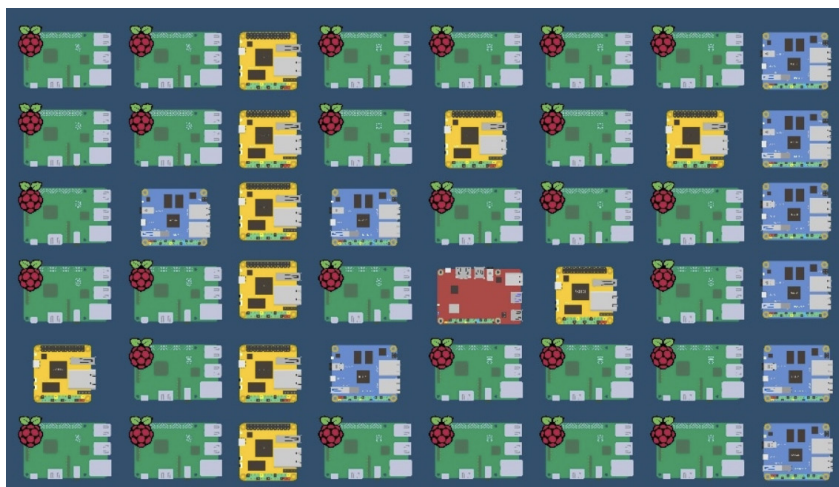


Figure 1: Scaling your IoT fleet

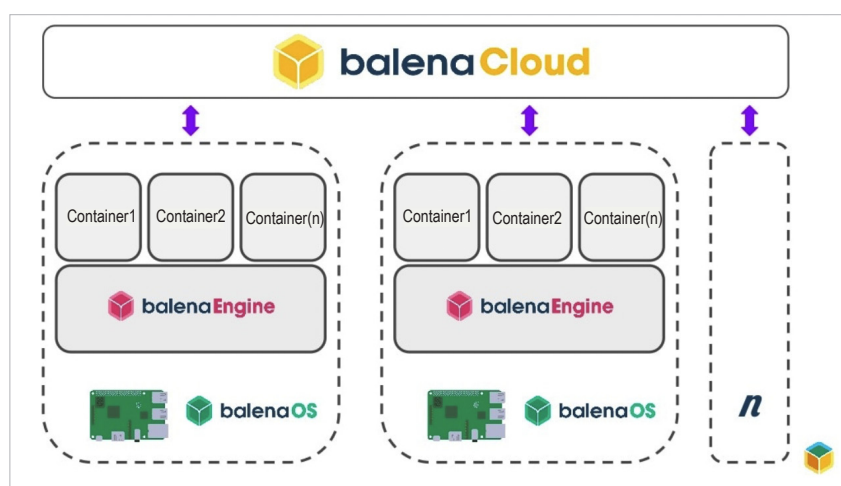


Figure 2: balena ecosystem

face. The balena ecosystem allows them to manage multiple edge devices straight from a single dashboard (balenaCloud), and adding a new device to an existing fleet is extremely straightforward. So you can focus on your main application rather than worrying about developing and scaling the infrastructure.

The balena ecosystem provides various tools that enable edge developers to adopt a container based Web-like development workflow, in which your application and supporting pieces reside inside Docker containers, managed by balenaEngine. The latter is similar to dockerEngine but suitably adapted for edge development. It runs on top of balenaOS, a Yocto based rugged

embedded Linux operating system meant for single board computers supporting ARM32, ARM64 and x86. A special container called supervisor, which comes baked into balenaOS, manages all the connection, provisioning and updation work from the device itself. A fleet owner can manage all the devices, add more and roll out an update release from balenaCloud, which provides a dashboard to monitor and manage all your devices in the fleet.

The balena ecosystem

Before diving into how you can get started with deploying your own fleet using balena, here's a quick recap of a few essential utilities.

- **balenaOS:** A Yocto based Linux operating system meant for edge devices; for example, Raspberry Pi
- **balenaEngine:** Container manager built into balenaOS, adapted for edge devices
- **balenaCloud:** Dashboard to monitor, manage, add and update devices in your fleet
- **balenaCLI:** A command line utility to push releases of your project to balenaCloud. balena also supports Git for this purpose, but for this tutorial we'll be using balenaCLI
- **balenaEtcher:** Some of you may already be using this great tool by balena for flashing SD cards/USB drives

It is a good time to mention that most (if not all) of these tools are open source and free to use; they are hosted on the balena GitHub account.

Note: You can add up to 10 edge devices for free, after which you need to buy the balenaCloud subscription.

Deploying your first IoT fleet with balena

We will now deploy our first IoT fleet with a sample 'hello-world' application. Here are the steps to do so.

Step 1: Create an account on balenaCloud

Step 2: Create a new fleet

Step 3: Add your device(s)

Step 4: Push your code to the devices in the fleet

Create an account: Visit balenaCloud (<https://www.balena.io/cloud/>) and create a free account. You can also log in using your Google or GitHub account.

Download balenaCLI: Download balenaCLI for your operating system from <https://github.com/balena-io/balena-cli/>.

Flash the SD card: Download balenaEtcher for your operating system to flash the SD card (from <https://www.balena.io/etcher/>), or you can use *build*